Question #: 1

A 13-year-old boy is brought to the physician by his mother because of a 2-day history of sore throat, runny nose and cough. His blood pressure is 90/70 mmHg, pulse is 90/min, respirations are 20/min and temperature is 38ºC (100ºF). Physical examination shows normal lung sounds. A diagnosis of an upper respiratory tract infection is made. Which of the following germ cell layers is the epithelial lining of the infected system most likely derived from?

✓A. Endoderm
B. Ectoderm
C. Paraxial mesoderm
D. Intermediate mesoderm
E. Lateral mesoderm

Rationale: The epithelial lining of the respiratory system is derived endoderm.

Question #: 2

A 56-year-old man is hospitalized following a car crash. Parenteral nutrition is started and dietary essential fatty acids of the omega-3 family are included in the supplements. Which of the following fatty acids most likely belongs to this group and is supplemented in this patient?

A. Palmitoleic acid
B. Oleic acid
C. Linoleic acid
✓D. Alpha-linolenic acid
E. Arachidonic acid
F. Palmitic acid
G. Stearic acid
H. Docosahexaenoic acid
I. Eicosapentaenoic acid

Rationale: Alpha-linolenic acid belongs to the omega-3 fatty acid family and is dietary essential. Linoleic acid is omega-6 and dietary essential fatty acid. DHA and EPA are omega-3 fatty acids; Arachidonic acid is omega-6 and not an essential fatty acid. Palmitic acid and stearic acid are saturated fatty acids

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Question #: 3

A researcher is analyzing the cell membrane of neuronal tissues and identifies that it has a high content of glycolipids with N-acetyl neuraminic acid (NANA) on the outer leaflet of the plasma membrane. Which of the following compounds most likely belongs to this class of lipids?

A. Phosphatidylethanolamine
B. Phosphatidylinositol
C. Sphingomyelin
✓D. Ganglioside
E. Triacylglycerol
F. Globosides
G. Cerebrosides
H. Cardiolipin

Rationale: Glycolipids containing NANA are gangliosides. Globosides and cerebrosides are also glycolipids but do not contain NANA. Phosphatidylethanolamine, phsophatidylinositol, cardiolipin are glycerophospholipids. Sphingomyelin is a sphingophospholipid. Triacylglycerol is a storage lipid and is not present in cell membrane.

Question #: 4

A 49-year-old man comes to the physician because of a 2-month history of skin lesions. Laboratory and clinical studies show stage IVB - cutaneous T-cell lymphoma. Vorinostat (Zolinza) which is an anticancer drug is prescribed to the patient. Vorinostat is a member of a large class of drugs that act as histone deacetylase inhibitors (HDAC inhibitors). Which of the following is the most likely mechanism of action of this drug?

A. Causes histones to become positively charged
✓B. Induces the formation of euchromatin
C. Decreases gene transcription
D. Increases the affinity of DNA to the nucleosome
E. Inhibits DNA methylation

Rationale: Vorinostat inhibits histone deacetylation, as a result the DNA is not wound tightly around histones. Acetylation of the histones results in increased expression of the genes and the formation of euchromatin.

Question #: 5

A 32-year-old woman comes to the physician because of a 4-day history of watery diarrhea following a week-long hiking trip. Physical examination shows diffuse abdominal tenderness and no evidence of peritoneal irritation. Stool examination for ova and parasites shows the presence of *Giardia intestinalis* cysts. This pathogen destroys the brush border of the intestines and the intestinal villi. Which of the following glucose transporters is most likely to be directly affected as a result of this infection?

A. GLUT-4
✓B. SGLT
C. GLUT-2
D. GLUT-1
E. GLUT-3

Rationale: SGLT-1 (Secondary active transporter for glucose) is present on the luminal surface of the intestinal mucosal cells.
GLUT-1 and GLUT-3 are present in brain and RBCs
GLUT-2 is present in the hepatocytes
GLUT-4 (insulin dependent) is present in adipocytes and muscle cells.
The GLUTs transport glucose by facilitated diffusion.

Question #: 6

Nasal epithelial cells are isolated from human nasal brushings and bathed in an extracellular solution containing: Na = 140 mmol, K = 5 mmol, Cl 105 mmol. Assume normal intracellular electrolyte concentrations of Na = 14 mmol, K = 140 mmol and Cl = 40 mmol. The membrane potential of one of the epithelial cells is measured at -60 mV. Assume that the constant RT/F = 60 mV. What is the driving force and direction of flux for sodium across the cell membrane?

✓A. 120 mV influx
B. 60 mV influx
C. 30 mV influx
D. 30 mV efflux
E. 60 mV efflux
F. 120 mV efflux

Rationale: The sodium gradient is 10x (140/14), making the Nernst potential for Na +60 mV. The actual Vm is -60 mV, and the driving force is the difference between Vm and the equilibrium potential for Na. -60

- 60 = -120 mV. Since the inside of the cell is negative, and the positive sodium ion has an inwardly directed concentration gradient as well as electrically attracted to the inside of the cell because of the interior negativity, an influx of ion into the cell results.

Question #: 7

A research team is developing a drug that targets a secondary active antiporter. Which of the following processes is most likely an example of secondary active anti-transport?

A. Glucose uptake into skeletal muscle
B. Transport of glucose from the intestinal lumen into an intestinal epithelial cell
C. Movement of the Na^+ ions into a nerve cell during the upstroke of an action potential
✓D. Na^+-dependent transport of Ca^{2+} from the cytosol to the extracellular fluid
E. Ouabain-sensitive transport of Na^+ ions from the cytosol to the extracellular fluid

Rationale: The Na-K ATPase is the main primary transport mechanism in most cells, directly using ATP to build a gradient of Na and K ions across the membrane. Secondary active transport mechanisms the use these gradients to build non-equilibrium gradients of other ion species, for example using the energy stored in the Na gradient to remove Calcium from the cell.

Question #: 8

Movement of Na^+ ions into a nerve cell during the upstroke of an action potential is being studied in a model system. Which of the following transport processes would most likely be observed during this study?

A. Primary active transport
B. Secondary co-transport
C. Secondary counter-transport
✓D. Facilitated diffusion
E. Passive diffusion through the cell membrane

Rationale: During the rapid depolarisation phase of a nerve action potential, voltage-sensitive Na+ channels open and allow the influx of Na+ ions into the cytosol. Ion movement is driven by the concentration gradient (Nernst potential), and the voltage-gated channels provide a facilitated diffusion route through the membrane, when open.

Question #: 9

A 25-year-old woman mountaineer comes to see her physician, explaining that she plans to climb Mount Everest. Her vital signs are within normal limits and physical examination shows no abnormalities. In preparation for the climb, she flies directly from sea level to base camp at an altitude of 5000 m. On reaching base camp, her ventilation rate increases as a result of the low PO2 activating her peripheral chemoreceptors. The increased ventilation then reduces her blood CO2 levels, which then trigger the central chemoreceptors to reduce ventilation rate. Safeguarding arterial oxygen is deemed more important than removing CO2, and her ventilation rate continued to be marginally elevated. What name is given to the homeostatic process outlined in this example?

✓A. Hierarchy
B. Redundancy
C. Feed forward regulation
D. Positive feedback
E. Competitive antagonism

Rationale: The homeostatic process described involves a hierarchical regulation, where one physiological need (maintaining oxygenation) is prioritized over another (removing CO2). In this scenario, safeguarding oxygen levels via increased ventilation takes precedence, even though this leads to a decrease in blood CO2 levels that would ordinarily reduce ventilation. The body essentially "chooses" to prioritize oxygenation over the effects of hypocapnia (low CO2), reflecting a hierarchical control mechanism.

Why not the other options?
b) Redundancy: Redundancy involves having multiple systems or pathways to maintain homeostasis. This situation does not involve redundant systems but rather a prioritization of one system over another.
c) Feed forward regulation: Feed forward regulation refers to mechanisms that anticipate changes and prepare the body in advance (e.g., salivating before eating). The process here is reactive to hypoxia, not anticipatory.
d) Positive feedback: Positive feedback amplifies changes rather than maintaining homeostasis (e.g., uterine contractions during labor). This example describes a homeostatic adjustment, not a self-reinforcing loop.
e) Competitive antagonism: Competitive antagonism refers to opposing molecules or pathways competing at the same site, which does not apply in this physiological context.

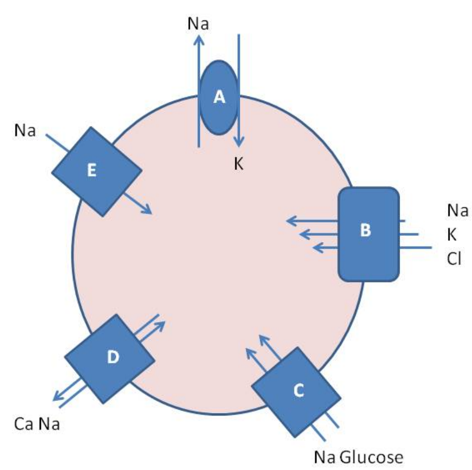
Question #: 10

A group of researchers is investigating primary active transport mechanisms in a cultured human cell line. The attached diagram depicts the active and expressed transport proteins in these cells. Which labelled transport process would be most immediately affected by uncoupling mitochondria and limiting ATP production?

✓A. A
B. B
C. C
D. D
E. E

Rationale: The Na/K-ATPase is the only example of primary active transport in the diagram. Operation of the transporter is directly driven by ATP hydrolysis, in the absence of ATP this process would be most immediately affected. The other transporters are passive or secondary active, which would continue to operate even without ATP until the driving ion currents have equilibrated across the membrane.

Attachment:



41132.jpg

Question #: 11

A 32-year-old woman in the third trimester of her pregnancy comes to the emergency room because of a 1-hour history of increasingly frequent and forceful uterine contractions. An increase in uterine muscle tension leads to the release of oxytocin from the pituitary gland. Oxytocin then stimulates a further increase in uterine muscle contractions and more tension develops. Which of the following homeostatic mechanisms best describes this phenomenon?

A. Negative Feedback
B. Redundancy
C. Feed-Forward Regulation
D. Hierarchy

✓E. Positive Feedback

Rationale: Understand the principles of positive and negative feedback in physiological systems. Positive feedback exacerbates the initial disturbance but with a definitive endpoint.

Question #: 12

A 24-year-old man is brought to the emergency room unconscious. The ambulance driver says on entering the ambulance, the patient was conscious, struggling to breathe, hypersalivating, sweating and demonstrating uncoordinated behavior. During the 15-minute journey to hospital he lost consciousness and started to seize. His wife says that in the restaurant he complained of numbness and tingling around his lips and extremities after consuming pufferfish. Pufferfish contains tetrodotoxin, a voltage-gated sodium channel blocker. His pulse is 95/min, blood pressure is 110/90 mm Hg and respirations are 5/min. He is immediately placed on oxygen after gastric lavage and given an IV bolus of alpha adrenoceptor agonist to increase his blood pressure. Which of the following pathophysiological findings is most likely seen in this patient's action potentials?

A. Complete absence of action potential formation
✓B. Slowed rate of depolarization
C. Delayed repolarization
D. Absence of repolarization
E. Slowed rate of repolarization

Rationale: The mention of pufferfish toxin (TTX) provides the best explanation for the symptoms in this otherwise healthy young man. Since TTX blocks voltage-gated Na channels, which are the key driver of depolarisation during the action potential, only answers A and B appear plausible, because all the other answers are concerned with repolarisation, which is mainly driven by potassium flux. The patient was still conscious when delivered to hospital, hence the dose of TTX ingested was not sufficient to block AP formation entirely. However, there is clearly an effect of the toxin (quite severe symptoms are being described) that disabled at least a portion of the voltage-gated Na channels, which would most strongly affect the rate of depolarization (answer B).

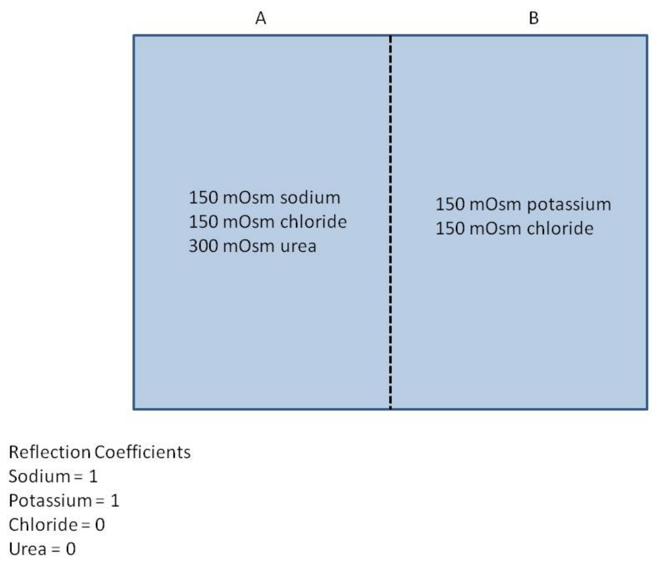
Question #: 13

Membrane properties are being studied in a setup where a semi-permeable membrane (dashed line) separates 2 compartments, as shown in the attached image. Which of the following best describes the tonicity and osmolarity of solution A compared to solution B, as illustrated in the attached image?

✓A. Isotonic and hyperosmotic
B. Hypertonic and hyperosmotic
C. Hypotonic and hyperosmotic
D. Isotonic and iso-osmotic
E. Hypertonic and hypo-osmotic

Rationale: Hyperosmolarity = all solutes, therefore, A = 600 mOsm; B = 300 mOsm. Thus, solution A is hyperosmotic to solution B.
Tonicity = all impermeant solutes with reflection coefficient =1, therefore both A and B = 150 mOsm. Thus, solution A is isotonic to solution B.

Attachment:



18836.jpg

Question #: 14

A healthy 21-year-old man is enrolled in an exercise physiology trial examining muscle fatigue. Physical examination is normal and shows him to be in excellent condition. Applied kinesiology and dynamic muscle function tests show normal nerve conduction and normal muscle strength at the beginning of the experiment. Which of the following mechanisms best explains the afterhyperpolarization of the motor action potentials observed in this man?

A. Increased sodium/potassium pump activity
B. Increased fractional sodium conductance
C. Increased potassium conductance
✓D. Increased fractional potassium conductance

E. Increased sodium conductance

Rationale: The hyperpolarisation after an action potential is due to the increased opening of voltage-gated (delayed rectifier) potassium channels, whilst the voltage-gated sodium channels are already closing. The changing conductances for both ions lead to a transient increase in the fractional conductance for potassium, which brings the membrane potential closer to the equilibrium potential for potassium, causing a transient hyperpolarisation of the cell membrane.

Question #: 15

A 23 year-old man comes to the physician 2 hours after a marathon run on a hot day because he is feeling very faint and dizzy. Isotonic fluid replacement is considered to be the best course of treatment. Assuming standard cellular concentrations and osmolarity (300 mOsm), which of the following solutions would most likely be isotonic?

✓A. 150 mM NaCl + 20 mM Urea
B. 300 mM NaCl
C. 150 mM NaCl + 20 mM dextrose
D. 300 mM NaCl + 20 mM Urea
E. 300 mM NaCl + 20 mM dextrose

Rationale: NaCl breaks down into Na+ and Cl- ions in solution, so it's osmolarity is twice the NaCl concentration. Reflection coefficient is 1, hence 150 mM NaCl yields 300 mOsm of tonicity. Urea can pass through membranes, i.e. has a reflection coefficient of 0, so it does not contribute to tonicity, which means that the 150 mM NaCl + 20 mM urea solution would be isotonic. Dextrose cannot pass the membrane passively, hence the 150 mM NaCl + 20 mM dextrose solution would be hypertonic.

Question #: 16

A 15-year-old boy is brought to the physician because of a 2-day history of muscle weakness and fatigue, especially after exercise. He says his limbs feel 'like rubber'. Laboratory studies of blood electrolyte levels are shown in the table below. Using the values in table, which of the following values best approximates the electrochemical equilibrium potential of potassium in this patient?

Ion	extracellular (mM)	intracellular (mM)
Na^+	140	14
K^+	2.4	120
Cl^-	104	4
Ca^{2+}	2.1	10^{-6}

A. -120 mV
✓B. -102 mV
C. -88 mV
D. -71 mV
E. -54 mV

Rationale: Serum K is very low, the patient has hypokalemia. The larger K gradient compared to normal (4.5/120 vs. 2.4/120) means that the Nernst potential is greater than usual, so it must be more negative than -90 mV. The correct answer is --60 * log(50) = 102 mV. Remember that the absolute value of the number is the potential magnitude, the sign only indicates a direction of the gradient.

Question #: 17

A 16-year-old boy is brought to the emergency room after being unable to walk and struggling to stand without help after competing at an athletics event. He says he feels extremely weak. He has a family history of primary hypokalemic periodic paralysis. Laboratory studies show that his plasma potassium level is 2.1 mmol/L (normal range 3.5 –5 mmol/L). He is treated with a K^+ supplement in a saline IV infusion. Which of the following is the most likely primary cause of his weakness?

A. Depolarized rest membrane potential of excitable cells
✓B. Hyperpolarized resting membrane potential of excitable cells
C. Increased refractory period of voltage gated sodium channels
D. Depolarizing blockade of voltage gated sodium channels
E. Increased opening of delayed rectifier potassium channels

Rationale: At this level of severe hypokalemia the Nernst equilibrium potential for potassium is significantly lowered (log(140/2.7) * -60 = -102 mV), which causes hyperpolarization of the membrane. This is the correct answer, depolarization would be the opposite. Refractory periods would decrease if there is any change. Voltage-gated sodium would be less likely to be blocked (in refractory state), and voltage-gated potassium channels would be less likely to be open.

Question #: 18

A 22-year-old man comes to the physician because of a 2-day history of diarrhea. His blood pressure is 128/86 mm Hg, pulse is 64/min and respirations are 16/min. Increasing ventilation of the lungs will result in lower plasma CO_2 concentrations, making the blood more alkaline and suppressing breathing. Conversely, decreasing ventilation will result in higher plasma CO_2 levels, causing increased acidity and stimulating breathing. This homeostatic regulation is best described by which of the following mechanisms?

- ✓A. Negative feedback
- B. Positive feedback
- C. Feed-forward regulation
- D. Redundancy
- E. Change of homeostatic setpoint

Rationale: This a system with classical negative feedback loops: ventilation above set-point for plasma CO2 levels gives a negative feedback signal to respiration, which disappears when ventilation drops below levels required to maintain normal plasma Co2 levels.

Question #: 19

A 56-year-old man comes to the emergency department because of a 2-week history of generalized weakness, dizziness, and fatigue and a 1-day history of palpitations and fainting. His blood pressure is 90/60 mmHg and pulse is 140/min and irregular. Electrocardiogram shows atrial fibrillation. The physician terminates the atrial fibrillation in the emergency department, and the patient is discharged on digoxin. This medication is known to inhibit the sodium-potassium pump found in the cell's plasma membrane. Which of the following constituents of the cell's membrane is most likely affected in this case?

- A. Glycolipids
- B. Cholesterol
- ✓C. Integral proteins
- D. Phospholipid
- E. Peripheral proteins

Rationale: The cell's membrane is composed of lipids and proteins. The 3 types of lipids are phospholipids (most abundant), glycolipids and cholesterol. Cholesterol is responsible for the stability of the membrane. The proteins component includes integral membrane proteins and peripheral membrane proteins. The sodium-potassium pump is an example of an integral membrane.

Question #: 20

A 40-year-old woman comes to the physician because of a 3-month history of a swelling on the medial aspect of her arm. Physical examination shows a nontender, mobile mass in the same area. The physician suspects a lipoma, a benign tumor of adipocytes. A biopsy is obtained for histopathological examination. Which of the following can be used as a fixative and stain to preserve and visualize the lipid in these cells for microscopic study?

A. Hematoxylin
B. Eosin
C. Silver
D. Periodic acid Schiff
✓E. Osmium tetroxide

Rationale: Osmium tetroxide is used as a secondary fixative to fix and visualize fats in cells and tissues. Hematoxylin, eosin, PAS and silver stains are never used as a fixative.

Question #: 21

A 63-year-old woman comes to the physician for a routine colonoscopy. During the procedure a mass is observed in the ascending colon. A biopsy is taken and a karyotype shows aneuploidy within the cells taken from the mass. The epithelium of the affected structure is usually a single cell layer thick with cells that are taller than they are wide. Which of the following best characterizes this epithelium?

✓A. Simple columnar
B. Transitional
C. Pseudostratified ciliated
D. Stratified cuboidal
E. Stratified squamous

Rationale: Aneuploidy is the abnormal number of chromosomes in a cell. Aneuploidy can originate during cell division when chromosomes do not separate properly & is regularly observed in malignant tumors.

The epithelium described here is that of simple columnar. Simple - single layer of cells.
Columnar - cells are taller than they are wide.

Transitional: epithelium lining the lower urinary tract, it is a stratified epithelium, with specific morphologic characteristics that allow it to distend.
Pseudostratified ciliated: appears stratified, although some of the cells do not reach the free surface, all rest of the basement membrane.
Stratified cuboidal: more than one layer in the epithelium, but the most apical layer, the width, depth and

height of the cells are approximately the same.
Stratified squamous: more than one layer in the epithelium, but the most apical layer, the width of the cell is greater than its height.

Question #: 22

A 32-year-old woman comes to the physician because of a 1-week history of fluid-filled blisters on her upper limbs. Electron microscopy of biopsy of these areas, show the separation of epithelial cells from the basement membrane. Which of the following cell adhesion molecules is a component of the junctional complexes found in this region?

A. Selectin
B. Cadherin
✓C. Integrin
D. Desmoglein
E. Claudin

Rationale: Hemidesmosomes are strong anchoring junctions located on the basal surface of epithelial cells. Integrins are the main family of proteins involved in hemidesmosomes that bind to extracellular glycoproteins such as laminin.
Selectins are CAMs that are expressed on leukocytes and endothelial cells and mediate neutrophil-endothelial cell recognition. They also aid in lymphocytic homing.
Cadherin: are represented by the family of transmembrane calcium-dependent CAMs localized mainly within the zonula adherens.
Desmoglein: helps with the linkage between the plasma membranes of adjacent cells by being part of the macula adherens/desmosomes.
Claudin: a transmembrane protein found in zonula occludens/tight junctions.

Question #: 23

A transmission electron microscopic study of endothelial cells of the lung vasculature involved in gas exchange is shown in the attached image. Which of the following mechanisms best describes the formation of these small, smooth vesicles (arrows) in the cytoplasm of these cells?

A. Autophagy
B. Exocytosis

C. Phagocytosis
✓D. Pinocytosis
E. Receptor-mediated endocytosis

Rationale: Pinocytosis (cell drinking) is the nonspecific ingestion of fluid and small molecules via small, smooth vesicles. Pinocytosis is performed constitutively by most cells and the smooth vesicles are especially numerous in the endothelium of blood vessels.
Autophagy: the process by which cell organelles and other cell components are transported to lysosomes and degraded.
Exocytosis: process by which a vesicle moves from the cytoplasm to the plasma membrane where it discharges its contents to the extracellular space.
Phagocytosis: ingestion of large particles such as cell debris, bacteria and other foreign materials.
Receptor-mediated endocytosis: allows entry of specific/selected molecules into the cell.
Attachment:

Question 7.jpg

Question #: 24

A biomedical scientist is investigating the effect of a tumor supressant on an aggressive tumor. The drug is found to inhibit the function of cells which have features similar to the cell in the attached photomicrograph. Which of the following functions will the drug most likely be inhibiting?

A. Steroid hormone production
B. Phagocytosis
✓C. Enzyme secretion
D. Contraction
E. Lipid storage

Rationale: The predominant organelle in this cell is Rough Endoplasmic reticulum indicating protein synthesis/ secretion is its main function. Enzymes, by their nature are proteins.

Attachment:

secretory cell 1.png

Question #: 25

A biotech company is conducting research on a new compound that functions as a chemotactic agent. Cell migration experiments show that neutrophils form lamellipodia and move towards the new compound in cell culture. Which of the following is most likely responsible for the formation of this cellular protrusion?

✓A. Actin polymerization
B. Intermediate filament polymerization
C. Kinesin
D. Microtubule polymerization
E. Mysosin

Rationale: Cell motility and the formation of cellular protrusions are driven by actin polymerization.

Question #: 26

A 21-year-old woman comes to the physician because of a 6-month history of severe acne on her face. Physical examination shows some pimples are acutely inflammed and a few have some pus inside, which can be easily expressed out.The sebaceous glands involved in this condition have a secretory process wherein the entire cell becomes a part of the secretory product. Which of the following best describes this mode of secretion?

A. Merocrine
✓B. Holocrine
C. Autocrine
D. Paracrine
E. Apocrine

Rationale: Different types of exocrine glands can exhibit different mechanisms of secretion. In holocrine secretion, the secretory products accumulate in the maturing cells. The cells then undergo a type of programmed cell death releasing both the secretory products and cellular debris.
Merocrine secretion - vesicles from within the cytoplasm fuses with the apical plasma membrane and gets released into the extracellular space.
Apocrine secretion - the vesicles accumulate at the most apical part of the cell and the apical part is eventually pinched off with those vesicles within it to release its contents to the extracellular space.
Autocrine and paracrine modes of secretion are seen in endocrine glands, not exocrine glands.

Question #: 27

An 8-year-old boy is brought to the physician by his mother because of a 3-day history of fever and a cough productive of greenish sputum. He is diagnosed with pneumonia (microbial infection of the respiratory passages). This is his 3rd episode of pneumonia in the last 6 months. Further testing shows that he has a dysfunction of the apical specialization cilia. Which of the following best characterizes the function of the affected structure?

A. Increases the surface area for absorption
B. Barrier against infections
C. Compartmentalization of mitochondria
✓D. Movement of trapped particles along surfaces
E. Adhesion to neighboring cells
F. Secretion of mucus

Rationale: This condition is associated with motile cilia. Cliia is used for motility. In this condition, he is unable to move particles and mucus along the surface of the epithelial cells. Hence the recurrent pneumonia.

Question #: 28

A 32-year-old woman comes to the physician for a follow up examination. She received a stab wound in her abdomen 3 weeks ago. Physical examination today shows that the sides of the wound are progressing well toward each other. The physician notes that this action is contributed to by the influence of a certain junctional complex that links the cell's actin cytoskeleton to the extracellular matrix. Which of the following transmembrane protein is a component of this junctional complex?

A. Selectin
B. Cadherin
✓C. Integrin
D. Desmocolllins
E. Claudin

Rationale: Both focal adhesions and hemidesmosomes are strong anchoring junctions located on the basal surface of epithelial cells. Integrins are the main family of proteins involved in both focal adhesions and hemidesmosomes that bind to extracellular glycoproteins such as fibronectin and laminin respectively.

Question #: 29

A pharmaceutical company discovers a new compound that inhibits the release of a secreted product that is continuously delivered to the plasma membrane for export. Which of the following processes is most likely being inhibited by this compound?

A. Phagocytosis
B. Pinocytosis
C. Exocytosis, regulated pathway
✓D. Exocytosis, constitutive pathway
E. Receptor-mediated endocytosis

Rationale: Constitutive pathway for exocytosis is utilized for exporting substances that are continuously delivered to the plasma membrane.

Question #: 30

A 26-year-old man is brought to the emergency department because of a 30-minute history of shivering following the use of an illegal drug. His blood pressure is 90/40 mmHg, pulse is 123/min, and respirations are 21/min. Physical examination shows a cold, sweaty, and almost unconscious young man. Treatment of this patient requires knowledge of the presence of cytochrome P450 within liver cells (hepatocytes). In which of the following organelles within the hepatocyte is this enzyme most likely located?

A. Lysosomes
B. Peroxisomes
✓C. Smooth endoplasmic reticulum
D. Proteasomes
E. Golgi apparatus

Rationale: The sER is particularly well developed in the liver and is the principal organelle involved in the detoxification of noxious substances.

Lysosomes: Digestive organelles, function in the controlled intracellular digestion of macromolecules.

Peroxisomes: Peroxisomes (microbodies) are small (0.5 mm in diameter), membrane-limited spherical organelles that contain oxidative enzymes, particularly catalase and other peroxidases.

Proteasomes: which are protein complexes that enzymatically degrade damaged and unnecessary proteins into small polypeptides and amino acids

Golgi Apparatus: complex of flattened, membrane-enclosed cisternae. Functions include post-translational modification, sorting and packaging.